

Ankush Kaundal

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RESEARCH INTERESTS

Computational Science, Physics-Based Models for Subsurface Hydrology, Parameter Estimation Using Inverse Modeling for Subsurface Systems, Monte Carlo Simulations, Parallel Computing, Numerical Groundwater Modeling, Automation of Numerical Models, Billions of Inverse Simulations for Estimating Subsurface Model Parameters at Continental Scale, Surrogate/Hybrid Models (Physics-ML/Data-Driven) for Groundwater and Flood Modeling, Advanced Geospatial Data Analysis

EDUCATION

Doctor of Philosophy (Ph.D.)

2020 - Present

Department of Civil Engineering, Indian Institute of Science (IISc), Bengaluru, India.

Thesis: *Parameter Estimation and numerical modelling of groundwater in data-scarce regions.*

Advisor: Prof. Sekhar Muddu

CGPA: 9.0/10

M.Tech in Civil Engineering (Water Resources)

2018 - 2020

Indian Institute of Technology (IIT) Guwahati, Assam, India.

Thesis: *Analytical modelling of groundwater and subsurface drainage in sloping lands using homotopy perturbation method.* [[thesis link](#)]

Advisor: Prof. Gautam Barua

CGPA: 9.1/10

B.Tech in Civil Engineering

2013 - 2017

Jaypee University of Information Technology, Warknaghat, Himachal Pradesh, India.

Thesis : *Estimation of earth work in designing hill roads.*

Advisor: Prof. Ashish Kumar

CGPA: 7.7/10

INDUSTRY EMPLOYMENTS

Engineer II, AtkinsRéalis

May-2026 - present

- Leading the development of python based post-processing utilities and automation scripts for numerical models, while supporting the delivery of UK water and environmental projects. (Kent Lower Greensand, Wets Midlands Wharf, Mole Basin Model, London Basin, Bromsgrove)
- Development of numerical modelling tools for rapid anomaly detection, model diagnostics, and root-cause analysis.
- Numerical modelling of the London Basin under climate change and groundwater abstraction scenarios.

Consultant, Adani Greens/Fluxgen Technologies

Jun-2025 - Mar-2026

- Led a team of 6 to build a scalable lake-recharge numerical groundwater model.
- Developed a lake-recharge, transient numerical groundwater model using MODFLOW (packages: RCH, DRN, GHB, EVT, ETS) to assess flow patterns, flow budget and storage changes in aquifer.
- Using MODFLOW outputs to compute water budget of watershed (Storage Change, ET, GW-outflow,

Recharge, Runoff)

- Using PEST tool to estimate aquifer parameters like specific yield, hydraulic conductivity and seasonal recharge factors.

Consultant, *Smartterra*

Jan-2025 - Mar-2026

- Extreme Value Analysis of hydrometeorological extremes to derive design events for flood risk modelling.
- Development of hybrid/surrogate models combining physics-based HEC-HMS and HEC-RAS flood models with machine learning for accelerated flood prediction and risk assessment.
- Deep learning models to classify wetland vegetation in spectrally complex systems using hyperspectral satellite imagery.

TECHNICAL SKILLS

Programming Languages: Python, R, MATLAB, JavaScript.

Libraries: FloPy(MODFLOW), scikit-learn, Pytorch, TensorFlow, Selenium, multiprocessing, joblib, pandas, numpy, rasterio, pyproj, geopandas, SciPy, osgeo(gdal), matplotlib, seaborn.

Data Science: Parameter Estimation using Inverse modeling, Monte Carlo Simulations, Random Forests, Deep Learning, Optimization algorithms, Parallel Computing.

IDEs: Pycharm, VS Code, Spyder, Jupyter Lab, RStudio, Cursor.

Software Packages: HEC-RAS, HEC-HMS, MODFLOW 6, MODFLOW-2005 (GMS, ModelMuse), FEFLOW, SWAT, ArcGIS, QGIS, SURFER, AutoCAD, LibreCad.

Misc: Google Earth Engine, Linux, GitHub, GitLab, vim, bash, anaconda, Windows, MS Office.

PUBLICATIONS

1. **Kaundal, A.**, Muddu, S., 2026. *Sequential estimation of aquifer parameters using water table elevations.* Journal of Hydrology [[link](#)] [[pdf](#)]
2. High-frequency groundwater measurements capture bonus recharge in a humid tropical catchment. Journal of Hydrology (in-review) [[link](#)]
3. Kaundal, A. and Muddu, S.: A simplified approach to modelling groundwater dynamics in complex, data-scarce semi-arid basins, EGU General Assembly 2025, Vienna, Austria. [[link](#)] [[pdf](#)]
4. Approaches to model groundwater in data-scarce basins using multiple data sources. (prepared)

INDUSTRY PROJECTS

- Flood modelling using HEC-RAS/HEC-HMS, flood hazard mapping and risk assessment for a vehicle manufacturing plant in Pantnagar, Uttarakhand. Client suffered huge business losses due to floods in Oct 2021. A flood model was developed to identify the most critical plant units vulnerable to flood risk. A detailed analysis was conducted, and the report was prepared and submitted.
Client: **TATA Motors** Oct-2022
- Groundwater modeling using MODFLOW for rainwater harvesting and artificial recharge through lakes. Built transient model using GHB, RCH, DRN packages to estimate the aquifer recharge through lakes. Used PEST tool to calibrate the model. Prepared and submitted scientific report to client.
Client: **TATA Motors** Jun-2022
- Modflow Modelling of Pollutant transport of effluents of paper mill on an irrigated agriculture.
Client: **Andhra Papers Limited** Nov-2023

- Water supply scheme and water treatment plant design. (Structural & environmental design, DPR preparation)
Client: **Mastiff Hotels company** Oct-2021

ACADEMIC PROJECTS

- Modelling sustainable groundwater extraction using numerical models, optimization and sequential parameter estimation to estimate pumping for Bengaluru International Airport (BIAL).
Designation: Researcher 2024-2025
- Modelling groundwater in cauvery river basin, Ministry of Earth Sciences Project (MoES).
Designation: Researcher 2021-2023
- A feasibility study on aquifer storage of groundwater by artificial recharge in and around Punjab region, India.
Designation: Research Fellow 2020

CONFERENCES/WORKSHOPS

1. **European General Assembly (EGU) 2025.**
Presented a poster on "A simplified approach to model groundwater dynamics in data-scarce regions"
Venue : Vienna, Austria Apr-2025
2. **Indo-French Monsoon School on Hydrology from Space.**
Venue : IISc Bengaluru, India Oct-2023
3. **Advanced Data-driven and Artificial Intelligence Tools in Water Resources (AI4Water).**
A short course on applying machine learning techniques in hydrology.
Venue : IIT Roorkee, India Aug-2023
4. **Workshop on Reference Evapotranspiration and Crop Coefficients.**
Workshop by Indo French cell, IISc to partition evaporation and transpiration
Venue : IISc Bengaluru, India Apr-2023
5. **International Soil Moisture School (IEEE GRSS).**
Venue : IIT Bombay, India Mar-2023
6. **Modelling groundwater flow and quality.**
MIKE powered DHI training course, Kochi India.
Venue : Kochi, India Jan-2023
7. **Terrestrial Modelling and High-Performance Scientific Computing.**
HPSC TerrSys Fall School - Coupled large scale high resolution models
Venue : University of Bonn, Bonn, Germany Oct-2022

ACHIEVEMENTS

- Received a full-time offer for the **Nat Cat Specialist/Modeller** position in **Swiss Re's** global **Risk Scientists** team. 2026
- Received full-time offers from three startups—**SmartTerra**, **FluxGen Technologies**, and **Clearwater Dynamics**—for roles involving numerical groundwater modelling and hybrid physics-ML modeling. 2026
- Independently delivered numerical groundwater modelling consultancy projects for **Tata Motors** and **Adani Green Energy**.
- **Mukhyamantri Protsahan Yojna**—Scholarship by the Government of Himachal Pradesh. 2018
- **Graduate Aptitude Test in Engineering (GATE 2018)**—Secured 99 percentile and received the Ministry of Human Resource Development (MHRD) scholarship.

TEACHING/COURSE ASSISTANT

- Groundwater Hydrology, CE 214, IISc Bengaluru: Taught groundwater modelling using MODFLOW; developed example simulations/assignments for class.
- Fluid Mechanics, CE 217, IISc Bengaluru
- Fluid Mechanics, CE 203, IIT Guwahati

POSITIONS OF RESPONSIBILITY

- Vice President (Finance) of **Entrepreneurship and Innovation cell of IISc Bengaluru**.
 - Core Member – **Red Ribbon Club, IIT Guwahati**.
 - Core Member – **Media and Publicity, CEC JUIT Waknaghat**.
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